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Design, implementation, and testing of a single axis levitation system for the suspension of a platform

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Abstract

This paper describes the design and implementation of a single axis DC attraction type suspension system, where a platform (vehicle structure) of around 14 kg mass is made to remain suspended at the desired operating gap under a ferromagnetic guide-way. The prototype has four electromagnetic actuators of attraction type and four inductive gap sensors, all located at the corners of the platform. The four actuators are controlled independently through four identical controllers, and the stable levitation of the platform is achieved through the *single input and single output* (SISO) control of each air-gap. The emphasis of this work is on the design and development of the switched mode power amplifier cum controller unit for the four actuators. The proposed single switch-based power circuit simplifies the overall hardware, and it can be extended to any number of magnet-coils. A cascade lead compensation control scheme utilizing an inner current loop and outer position loop has been designed and implemented for the stabilization of such a highly unstable and strongly nonlinear system. The prototype has been successfully tested, and stable levitation was demonstrated with the desired operating gap.

Keywords: Electromagnetic levitation; Multi-magnet based levitation; SISO control; Switched mode power amplifier; Lead compensation

1. Introduction

Most of the reported literatures [1–5] in the field of electromagnetic levitation are about simple single magnet based systems. Though these single magnet, single axis levitation schemes may be useful for some industrial applications, the majority of applications require a multi-axis levitation control, where one may need to use a multiple-magnet based levitation system. This part of the work reports on the control of an electromagnetically levitated platform (vehicle structure) that uses four attraction type magnets fixed at the four corners of the platform. The overall structure of the electromagnetically levitated vehicle may consist of independent levitation, guidance, and propulsion systems. However, this work considers only the singe axis magnetic levitation part. Basically the concepts and principles of single actuator based levitation [5] have been extended to control all the four magnets. The four electromagnets are fabricated so as to have identical electrical and mechanical parameters. Each of these four magnets is fitted at one corner of the platform, and is controlled to maintain the required air-gap of its respective corner. One inductive type position sensor is mounted near each corner, and the corresponding magnet (actuator) is independently controlled to maintain the desired air-gap. By applying suitable vertical gap commands to the four electromagnets, the roll, pitch, and heave can also be controlled to some extent. However, for achieving better dynamic characteristics of the suspended platform, independent control of roll, pitch, and heave movement is required [1]. In this integrated control approach [1], the control loops are non-interacting, and thus may be designed for each degree of freedom without any reference to cross-coupling. One inductive-type position sensor is mounted near each corner, and the corresponding magnet (actuator) is independently

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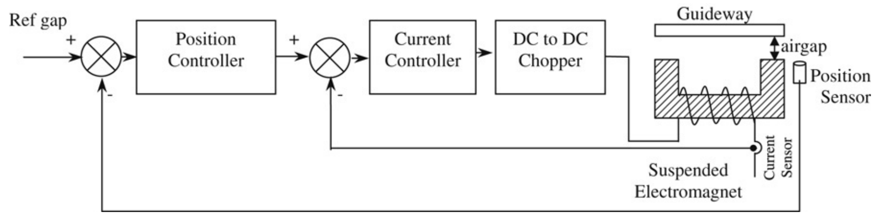

Fig. 1. Block diagram of the proposed scheme.

controlled to maintain the desired air-gap. The DC EMLS is highly nonlinear in nature due to the inherent non-linear force–distance–current characteristics of the electromagnet. After making appropriate assumptions, the non-linear equations are linearised about an operating point and a linear model has been derived. Since DC EMLS is an inherently unstable system, the stabilization of the overall closed loop system is one of the main objectives of this work. A cascade-lead compensation control [9] scheme utilizing an inner current loop and an outer position loop has been developed. Controllers have been designed by utilizing classical root-locus and frequency domain techniques. The four controllers for the four magnets are almost identical. The block diagram of each controller unit is shown in Fig. 1. The overall hardware gets considerably simplified due to the use of the proposed single switch based power converter circuit [7]. Detailed descriptions of the experimental setup, modelling, design of the controller, hardware circuits, and simulated and practical results are presented in the following sections.

2. Experimental setup

The prototype (total mass 13.84 kg) consists of four identical electromagnets placed at the four corners of a platform (Fig. 2), and the structure is made to remain suspended at different air-gap positions under a ferromagnetic guide-way — the arrangement that is normally used for an electromagnetic MAGLEV system. The minimum number of actuators required to levitate such a platform is four, but more than four actuators may be used for better reliability [6]. The details of each magnet are given in Table 1. In this work, a U-core type magnet and flat guide-way have been chosen, for a better lift/drag ratio [3]. The laminated rail and magnet-core is advantageous not only in reducing the magnetic drag force caused by eddy currents in the iron rail, but also to make a quicker control response of the air-gap between the magnet and the rail. For simplicity and rigidity in the structure, a solid iron rail and core have been used. The platform is made of a bakelite sheet of 10 mm thickness. The guide-ways through which the magnetic flux flows are constructed with two solid iron channels. Four identical inductive-type position sensors are fitted with the help of aluminium plates at the four corners of the fixed (guide-way) structure to measure the air-gap information of each corner. The stable suspension of the prototype is achieved through the control of the air-gap between the magnet and the guide-way by controlling the current flowing through the magnet-coil. The four magnets are controlled independently through four identical controllers, and stable levitation of the platform is achieved through single input and single output (SISO) control of each air-gap. In each case, the current of the electromagnet is controlled through the chopper circuit, utilizing an outer position control loop and an inner current feedback control loop as shown in the block diagram of Fig. 1. With each magnet cum controller unit working satisfactorily, each corner of the vehicle gets the desired vertical lift, and in the process the whole MAGLEV structure is levitated. To stop the drifting of the vehicle laterally (in the horizontal plane), some amount of saliency is provided in the iron face of the steel girder (Fig. 2). The pole faces of the magnets tend to remain vertically below the salient face of the steel girder so as to have the least reluctance path for the magnetic flux.

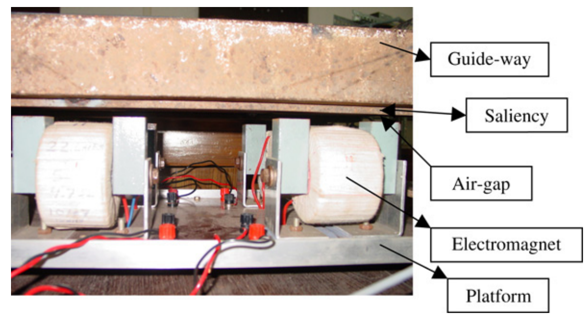

Fig. 2. Photograph of the experimental set up (front view).

Table 1
Parameters of the magnet-coil

Mass of the electromagnet	3.051 kg
No of turns of the coil	1040
Gauge of enamel copper wire	19
Resistance of the coil	5.50 Ω
Inductance of the coil	0.200 H

3. System modelling and controller design

The four magnet-coils used in this work are identical to each other, having almost the same mechanical and electrical parameters. Assuming a uniform distribution of the load, the upward lift force to be generated by each magnet is equivalent to 3.46 kg, and this leads to the magnet parameters being derived by the technique used in single magnet levitation.

Many researchers have tried to analyse and control the electromagnetic levitation system based on a linear approximation of the nonlinear model. Most researchers have taken voltage as the exciting input to the magnet-coil, and this results in a third order unstable system [3]. In this approach, the magnet-coil has been excited by a controlled current source, which results in a simpler second order transfer function for the system.

Assuming all the flux generated by the electromagnet passes through the ferromagnetic guide-way, the instantaneous coil inductance may be expressed as:

$$L(z) = \frac{N}{i(t)}\Phi_T = \frac{N^2}{R_T} \tag{1}$$

where, $N =$ No of turns of the coil,
$i(t) =$ Instantaneous current through the coil,
$\Phi_T =$ Total flux in the magnetic circuit,
$R_T =$ Reluctance of the entire magnetic circuit.

If the reluctance of the magnetic core is assumed to be negligible when compared to the two air gaps, we have [total length $= 2z(t)$],

$$L(z) = \frac{\mu_0 N^2 A}{2z(t)}. \tag{2}$$

At any instant of time, the force of attraction between the electromagnet and the ferromagnetic rail is given by

$$F(i,z) = -\frac{\mathrm{d}}{\mathrm{d}z}\left[\frac{1}{2}L(z)i(t)^2\right]. \tag{3}$$

Now putting the inductance value from Eq. (2) into the force equation (3), one can write:

$$F(i,z) = \frac{\mu_0 N^2 A}{4}\left[\frac{i(t)}{z(t)}\right]^2. \tag{4}$$

So, at the equilibrium position (i_0, z_0), the normalized force equation is

$$F_0(i_0, z_0) = \frac{\mu_0 N^2 A}{4}\left[\frac{i_0}{z_0}\right]^2 = mg. \tag{5}$$

The dynamics of the electromagnet are given by the following equations

$$\begin{aligned} m\ddot{z}(t) &= -F(i,z) + mg; \\ &= -\frac{\mu_0 N^2 A}{4}\left[\frac{i(t)}{z(t)}\right]^2 + mg. \end{aligned} \tag{6}$$

The system dynamic equations are thus nonlinear and hence difficult to analyse. So the equations are linearised about a suitable operating point (i_0, z_0), and the linearised model may be found as described below. If the mass of the electromagnet is displaced by an amount $\Delta z(t)$ from the stable point, then let the corresponding change in current be $\Delta i(t)$. The small perturbation linear equations (discounting second order effects) of the system are:

$$\begin{aligned} m\Delta\ddot{z}(t) &= \frac{\mu_0 N^2 A}{4}\left[\frac{i_0 + \Delta i(t)}{z_0 + \Delta z(t)}\right]^2 + mg \\ &= -\frac{\mu_0 N^2 A}{4}\left[\frac{i_0}{z_0}\right]^2\left[1 + \frac{2\Delta i(t)}{i_0} - \frac{2\Delta z(t)}{z_0}\right] + mg \\ &= -\frac{\mu_0 N^2 A}{4}\left[\frac{i_0}{z_0}\right]^2 - \frac{\mu_0 N^2 A i_0}{2z_0^2}\Delta i(t) \\ &\quad + \frac{\mu_0 N^2 A i_0^2}{2z_0^3}\Delta z(t) + mg \\ &= -K_a\,\Delta i(t) + K_z\,\Delta z(t); \end{aligned} \tag{7}$$

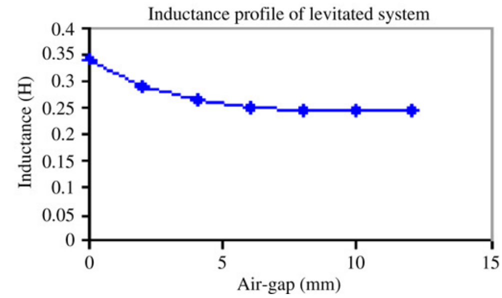

Fig. 3. Inductance profile of the levitated system.

where,

$$K_a = \frac{\mathrm{d}F(i,z)}{\mathrm{d}i} \quad \text{and} \quad K_z = \frac{\mathrm{d}F(i,z)}{\mathrm{d}z}. \tag{8}$$

So, these two force constants K_a and K_z are respectively the slope at the operating point of the force–current and the force–distance characteristics of the electromagnetic suspension system.

Taking Laplace transforms on both sides of Eq. (7) and after rearranging, the transfer function of the plant is written as:

$$G_p(s) = \frac{\Delta Z(s)}{\Delta I(s)} = -\frac{\left(\frac{K_a}{m}\right)}{\left(s^2 - \frac{K_z}{m}\right)}. \tag{9}$$

The negative sign in the expression indicates a decrease of the object's position with the incremental change of the coil-current or force. The inductance profile with distance for the levitated system, as found experimentally, is shown in Fig. 3. The inductance value around an operating point (in the medium gap range) mostly varies inversely with respect to the object (platform) position and has been approximated by Wong [4], as:

$$L(z) = L_c + \frac{L_0 z_0}{z} \tag{10}$$

where L_c is the inductance of the coil in the absence of the guide-way, and L_0 is the additional inductance contributed by the ferromagnetic guide-way.

Now the basic force equation becomes

$$F(z) = -\frac{i^2}{2}\frac{\mathrm{d}L(z)}{\mathrm{d}z} = C\left(\frac{i}{z}\right)^2 \tag{11}$$

where $C = \frac{L_0 z_0}{2}$ is a constant, which can be determined experimentally.

The partial derivatives of the force expression in Eq. (11) w.r.t. 'i' and 'z' (at the nominal current i_0 and air-gap z_0) will respectively be equal to K_a and K_z.

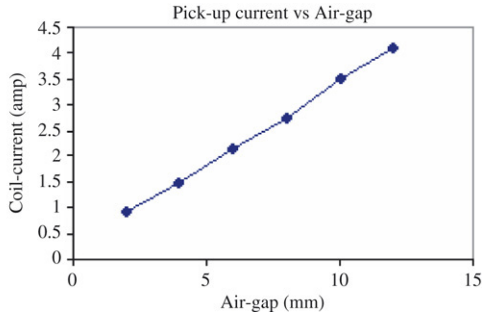

Fig. 4. Current profile of the levitated system.

Thus:

$$K_a = 2C\left(\frac{i_0}{z_0^2}\right) \quad \text{and} \quad K_z = 2C\left(\frac{i_0^2}{z_0^3}\right). \tag{12}$$

The instantaneous voltage across the magnet winding can be expressed as

$$v(t) = Ri(t) + \frac{\mathrm{d}}{\mathrm{d}t}\left[L(z)i(t)\right]. \tag{13}$$

It is assumed that when the system is properly designed, the suspended object will remain close to its equilibrium position, i.e., $z = z_0$, and from Eq. (10) the expression of inductance can be expressed as follows

$$L(z) = L_c + L_0 = L. \tag{14}$$

L is the total inductance of the coil at a particular gap position and is assumed to be constant at the operating point. So the voltage equation (13) can be written as

$$V(t) = Ri(t) + L\frac{\mathrm{d}i(t)}{\mathrm{d}t}. \tag{15}$$

Taking the Laplace transform of Eq. (15) and after rearranging, the model of the actuator can be written as below:

$$\frac{I(s)}{V(s)} = \frac{1}{(R + sL)}. \tag{16}$$

The pick-up current versus air-gap for the electromagnetic levitation system (EMLS) is shown in Fig. 4.

The selection of the operating point for the levitated object plays a significant role in the design and development of a DC EMLS. There are many reasons why the operating air-gap between the pole-face of the magnet and the object cannot be made arbitrarily too small or too large. In the higher operating gap zone, the required current for levitation is very high and the high value of the coil-current becomes a major constraint in the design of the actuator, power amplifier, and controller. The large magnitude of coil-inductance in the lower gap zone leads to a high (L/R) ratio, which consequently makes it more difficult to achieve faster control of the coil-current and levitation force. Considering all these points, the operating air-gap for the present system has been selected as 4 mm.

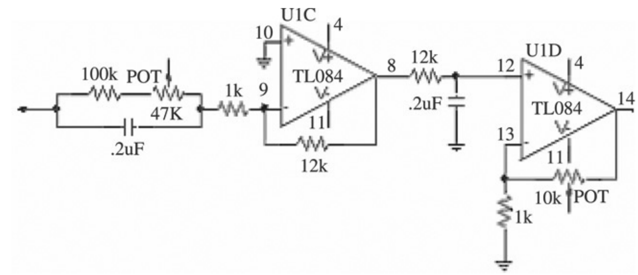

Fig. 5. Position controller details.

The transfer-function of the levitated system at an operating gap of 4 mm is given by

$$G_p(s) = \frac{\Delta Z(s)}{\Delta I(s)} = \frac{7}{(s+49)(s-49)}. \tag{17}$$

From Eq. (17), the plant $G_p(s)$ has one stable pole at $s = -49$ and an unstable pole at $s = 49$. The closed loop system has been stabilized by a cascade lead compensation technique [9] utilizing an inner current loop and an outer position control loop. The design of the control loop starts from the innermost (fastest) current loop and proceeds to the slowest position loop. The parameters of the current controller (PI controller) and position controller (lead compensator) have been designed based on the linear control theory utilizing the root-locus and frequency domain techniques. In the outer loop, the objective is to design a phase lead compensator so that the overall closed loop system becomes stable. The pole and zero are to be placed on the negative real axis, and there are many possible pole-zero locations for stabilizing the system. The phase-lead compensator design procedure in this case is to place the zero of the compensator between 0 and -49 on the real axis of the s-plane, while the pole of the compensator is placed nearly 10 times to the left of the zero position [11]. Simulation studies are carried out to find a pole-zero and to gain a combination that ensures the system is stable while giving acceptable performance. It must be mentioned that since the system is inherently unstable, in general, both stability and good performance cannot be achieved by using a single controller [10].

The transfer function of the designed lead controller for an operating air-gap of 4 mm is given by

$$G_{pc}(s) = \frac{9(s+47)}{(s+400)}. \tag{18}$$

The hardware realization of the position controller by an analog circuit is shown in Fig. 5.

The chopper amplifier can be modelled by a gain and a time constant which can be represented by the following transfer function.

$$G_{ch}(s) = \frac{K_c}{(1 + sT_c)}; \tag{19}$$

where K_c is the gain and T_c is the time constant of the chopper circuit.

With an input supply of around 100 V, the chopper gain (ratio between output voltage and control voltage coming from

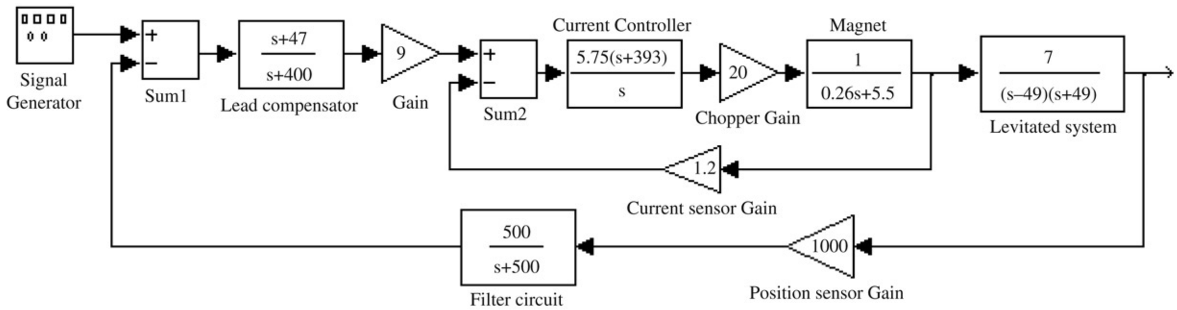

Fig. 6. Simulink (small signal) model of the overall closed loop system (one unit).

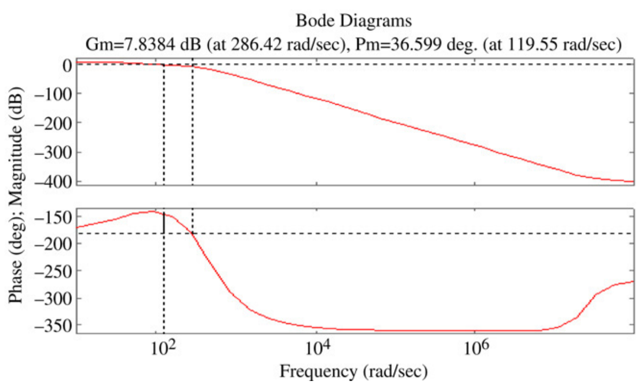

Fig. 7. Frequency response plot of the overall closed loop system.

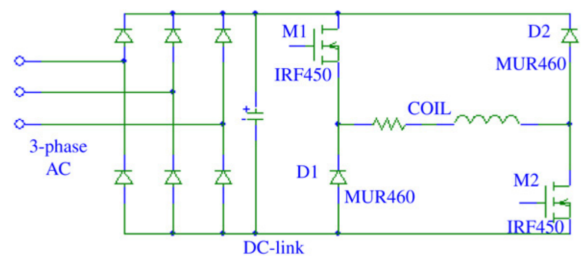

Fig. 8. Asymmetrical bridge (H-bridge) converter circuit.

current controller) will be equal to 20. The time lag in the converter is equal to the average carrier switching-cycle time, i.e. half the period, and is expressed in terms of the PWM switching frequency as $T_c = \frac{1}{2f_c}$. After neglecting the very small time constant for a very high switching frequency, the transfer function of the chopper becomes a simple gain of 20.

In the feedback path, a position sensor along with a low pass filter is used for the position measurement. The sensor has a good dynamic response (bandwidth of 1 kHz) with very little phase shift. The position sensor gain in SI units is calculated as 1000 V/m. A low pass filter circuit having a cut-off frequency of 80 Hz is used to remove the high frequency noise from the position signal before it is fed to the lead controller circuit. The transfer-function of the filter is given by the following form.

$$G_f(s) = \frac{500}{(s + 500)}. \tag{20}$$

An LEM current sensor (LA-55P) was used to measure the actual coil current for the individual unit. The current sensor block may also be approximated by a simple gain. In the case of a filtering requirement, a low-pass filter can be included in the analysis. Even then, the time constant of the filter might not be greater than a millisecond. The bandwidth of the sensor is very high (1 MHz), and there is perfect (for our purposes) linearity between the actual current and the sensed current signal.

The simulink (small signal) model of the overall closed-loop system (one unit) comprising the different elements is shown in Fig. 6. The frequency response plot of the overall system (Fig. 6) is shown in Fig. 7.

4. Power amplifier

In the present work, four magnets with their individual air-gap controllers have been used for levitating the rectangular platform. These four magnets have their separate power amplifiers and they are separately controlled. The asymmetric H-bridge converter (Fig. 8) is ideal for high power levitation systems, not only for its energy efficiency but also for its better dynamic performance (in the sense that the amplifier may be made to behave like a simple gain block). During experimentation, it has been observed that the actual current faithfully follows the reference current up to about 100 Hz (which is at least ten times faster than the position loop dynamics) for both step and sinusoidal reference inputs. For the four-magnet case, one needs to have four bridge circuits, each having two controlled switches and two diodes. Now, instead of an asymmetrical bridge circuit if the simplified circuit with only one switch (and a diode with an energy-dump circuit) per magnet is used, the overall saving in the power amplifier components may be significant. All the four switches of the simplified power circuit may be connected such that their gate drive signals may not need ohmic isolation. Moreover all four amplifiers may share a common energy dump capacitor–resistor circuit. The exact connection diagram is shown in Fig. 9. This circuit, shown in Fig. 9, uses half the number of switches and diodes used in the asymmetrical bridge circuit, and as a result the total EMI generation [8] is also reduced. The circuit is less energy-efficient compared to an asymmetrical bridge circuit, but more efficient compared to the linear amplifier. All such converters are required to achieve fast control over the magnetic current rise as well as the current fall. The magnet coil, being highly inductive, requires a sufficient magnitude of bi-directional voltage for quick control over the coil current. The asymmetrical bridge circuit (Fig. 8) applies a positive (supply) voltage to the coil when the two switches are simultaneously

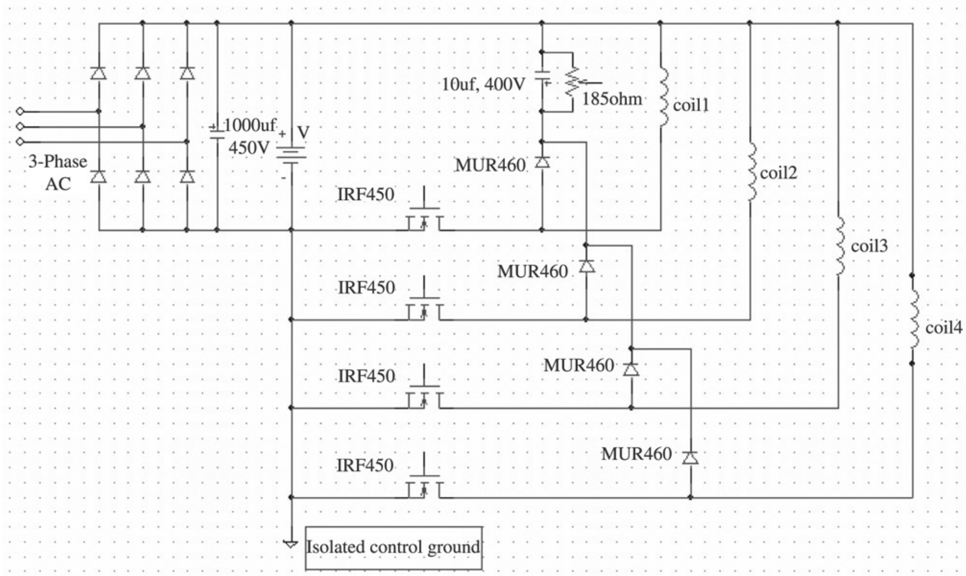

Fig. 9. Proposed switched mode power amplifier.

on, and applies a negative voltage when the switches are turned off, resulting in the conduction of the diagonally placed diodes. During the application of the positive voltage, the coil current rises fast and during the application of negative voltage the current decays fast. During this current decay, through the diodes, part of the magnet's energy is fed back to the supply, and thus the circuit is quite energy efficient. For stable suspension, one essential requirement of the power circuit is to quickly control the magnetic current for fast control over the electromagnetic lift force. Thus, not only the current build-up through the magnet coil needs to be fast, the current decay must also be quite fast. The simple single switch based power circuit (Fig. 9) achieves this objective. When the switch is on, the circuit applies a positive (supply) voltage to the coil. When the switch is off, the magnetic current, instead of flowing back to the supply, charges a capacitor. Under the steady operating cycle, the ripple voltage across the capacitor is reduced, and the magnet-coil is easily reverse-biased by the capacitor-voltage. To limit an indefinite rise in this capacitor voltage, a discharge resistor is put across it. The value of the discharge resistance can be predetermined knowing the circuit conditions. It may be seen here that the magnetic energy associated with current decay is wasted in the discharge resistor. However, the circuit, due to its simplicity and minimal noise production, may be quite attractive for low power magnet coils. With a four coil circuit, using a simplified chopper topology, it may make sense to invest in another simple circuit to control the dump capacitor voltage within desired limits, or to feed back this energy to the input supply. In spite of this additional modification, there is expected to be an overall saving in the power amplifier cost as compared to the four asymmetrical bridge circuit.

5. Experimental results

The present electromagnetic levitated system (four coil structure) has been successfully tested, and its stable levitation was demonstrated around the designed operating gap of 4 mm. The photograph of the experimental setup during stable levitation is shown in Fig. 10. Fig. 11 shows the four position signals at the levitating position. It is seen that the dc magnitudes of the position outputs do not vary much. A square wave disturbance (2 V, 2 Hz) has been applied to the air-gap reference of coil #4, and the system response has been shown in Fig. 12. From the experimental response, it is clear that the closed-loop system has been stabilized by the designed controller. Here, the controller designed for a fixed air-gap (4 mm) is fed with a steady gap command, superimposed with a repetitive step voltage of 2 V which corresponds to a 2 mm air-gap variation from the nominal operating point. The output voltage of an inductive type position sensor is 1 V for a 1 mm distance between the conductive material and the sensing surface. A steady-state error is noticed in the experimental response for the step input, because for this system the lead compensation cannot eliminate the steady-state error for a step input. This problem could be solved by introducing an integrator in the control loop, but it will increase the difficulty of designing a stabilizing controller for the system. Higher position error gain again will decrease the system damping, though it will reduce the position error to some extent. Actually, a single lead controller cannot satisfy both stability and performance requirements for such an unstable system [10]. To

Fig. 10. Photograph of the setup during stable levitation.

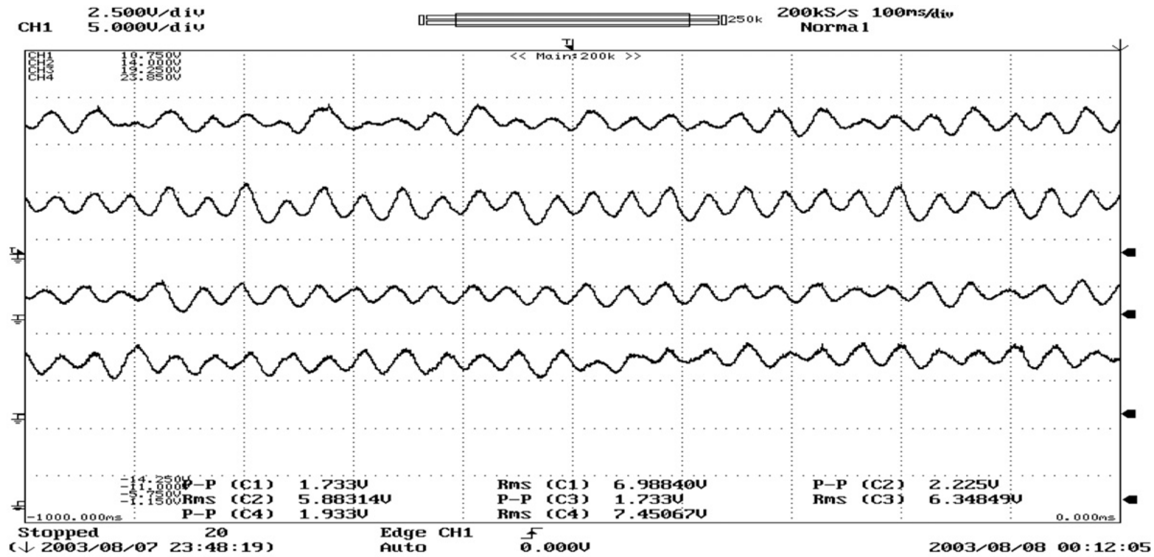

Fig. 11. Four position signals during levitation.

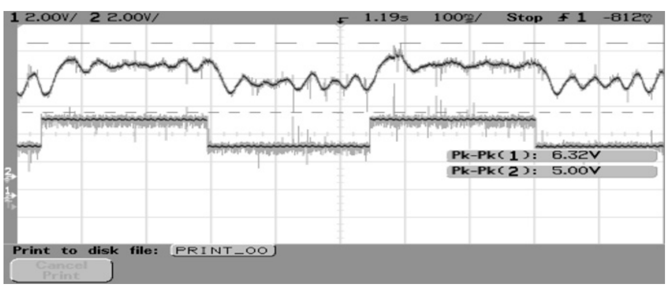

Fig. 12. Position response of air-gap#4 (CH2) due to square wave input (2 V, 2 Hz).

enhance the system's performance, another loop (along with a cascade PI controller) may be introduced into the two-loop structure of the EMLS presented in Fig. 6. There are a number of factors that degrade the overall performance of the system. Many factors, such as the sensitivity to parameter uncertainties, linear approximation of the electromagnetic force, unmodelled dynamics of the actuator, sensor noise, filtering time delay and amplifier bandwidth (frequency-band over which the power amplifier behaves as an ideal current source), cross-coupling between different units, etc. have a significant impact on the dynamic performance of the levitated system.

6. Conclusion

The four magnets have been controlled independently by four identical controllers, and the platform was levitated by employing an independent single input–single output control of each gap. The use of the decoupling technique simplified the overall analysis and design of the controller units. The cascade lead compensators for the four units have been designed by using the classical synthesis method. The four independent SISO controllers designed for controlling and maintaining a fixed air-gap at the four corners of the platform are found to work successfully in unison. The SISO controllers are able to rectify and compensate the errors introduced due to non-identical coils, and mechanical imperfections caused by in-house laboratory production. Though the four electromagnets are fabricated to have identical electrical and mechanical parameters, there might still be some variations in the parameters of the four coils due to many reasons. The inductance values may not be exactly equal at the actual operating point due to the non-uniformity of the air-gaps between the pole-faces of the magnets and the guide-ways. The weight sharing by the four magnets may also not be same. It has been assumed that there is an equal distribution of weight for each coil, but the coils may not be perfectly positioned to share the total weight equally. Also, the sharing of the weight-load during dynamic conditions may not be fixed. Due to all these factors, the system parameters for the four different units may not be the same during dynamic conditions. Moreover, in the present local control scheme, any disturbance entering in any loop or applied to the mechanics will cause a change to all variables and will have an effect on the other loops. For better stability, each controller has to be designed after considering the self and mutual cross-coupling of the different units. For better dynamic characteristics of the suspended platform, independent control of roll, pitch, and heave movement is required, and a separately roll, pitch, and heave controller has to be designed. To realize the appropriate feedback control laws, the implementation of the scheme would require more transducers to measure angles and angular velocities.

Due to the use of a *single* switch based switched mode power amplifier, the overall hardware circuit becomes simpler and more compact. The overall cost of the power amplifier as well as EMLS have also been reduced considerably by the use of this power circuit.

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